

Abstract

Reciprocal AI-Enabled Community-Engaged Learning in Environmental Data Analytics

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Artificial intelligence tools are increasingly integrated into higher education; however, there remains limited empirical research examining how AI-assisted analytical methods function within sustained, community-engaged experiential learning environments. This presentation introduces the Community-Engaged Earth Data and Sustainability (CEEDS) model, an AI-enabled instructional framework implemented in an undergraduate geoscience course in collaboration with local high school partners.

The CEEDS model builds upon prior experience from the NSF-funded GEOTeams program (IUSE #1911514), which employed multilevel research teams to broaden participation in geoscience education. Extending this foundation, CEEDS integrates RStudio programming, authentic environmental datasets (e.g., earthquake, groundwater, and climate data), and structured, reciprocal research collaboration between university students and high school students. Undergraduate participants receive training in AI-assisted data analysis and scientific communication before engaging in small mixed research teams that co-develop sustainability-focused projects and produce interactive dashboards and public presentations. Using a mixed-methods evaluation design, the project measures growth in AI literacy, programming competency, collaborative research confidence, and mentorship effectiveness through anonymous pre- and post-course surveys, rubric-based assessments, and coded qualitative reflections. Preliminary findings suggest that embedding AI tools within reciprocal research partnerships enhances student engagement and promotes deeper understanding of data-driven environmental analysis.

This presentation discusses pedagogical design principles, assessment strategies, and challenges associated with integrating AI within experiential and community-engaged learning contexts. The CEEDS model offers a replicable framework for institutions seeking to combine AI literacy, sustainability education, and cross-level collaboration in technology-enhanced teaching.

Keywords: Artificial Intelligence in Education, Experiential Learning, Community-Engaged Learning, STEM Education, Technology-Enhanced Pedagogy